

A Literature-Implied Answer To A Maurey-Type Operator-Space Problem

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Status

This is a **literature-implied answer**. No original proof is claimed in this packet. The role of the packet is to record that a later theorem answers the 2007 source problem after matching the source's dual reformulation with the later notation.

Source Problem

The source is Junge–Lee, *A Maurey type result for operator spaces*, arXiv:0707.0152 [1]. In the introduction, after defining completely p -summing maps, the authors ask whether for $2 < q < \infty$

$$CB(B(H), OH) = \Pi_q^o(B(H), OH).$$

They note that, if $p = q' \in (1, 2)$, this problem is equivalent to the Schatten-side dual formulation

$$S_p(OH) = \Pi_1^o(OH, S_p).$$

Remark 2.2 also states that the authors could not answer the related Schatten target inclusion

$$\Pi_p^o(OH, S_p) \subseteq \Pi_1^o(OH, S_p).$$

Supporting Theorem

The supporting source is Junge–Parcet, arXiv:0901.1928 [2]. Their Corollary B states, in particular, that for every $2 < s < \infty$ and every operator space Y ,

$$\Pi_{s'}^o(OH, Y) = \Pi_1^o(OH, Y).$$

Taking $s = q$, $p = s' = q'$, and $Y = S_p$, this gives

$$\Pi_p^o(OH, S_p) = \Pi_1^o(OH, S_p).$$

Together with the 2007 paper's stated dual equivalence, this supplies the desired answer to the source problem

$$CB(B(H), OH) = \Pi_q^o(B(H), OH) \quad (2 < q < \infty).$$

Provenance And Scope

This is not filed as `literature_already_answered`: the 2009 paper cites Junge–Lee and says its corollary vastly improves those results, but the exact 2007 question is not restated there as a named problem being resolved. The match uses the source paper’s own dual formulation and the later Corollary B. Hence the durable status is

`literature_implied_answer`.

References

- [1] M. Junge and H. H. Lee, *A Maurey type result for operator spaces*, J. Funct. Anal. 254 (2008), 1373–1409; arXiv:0707.0152.
- [2] M. Junge and J. Parcet, *Maurey’s factorization theory for operator spaces*, arXiv:0901.1928.